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16. Reverse Linked List

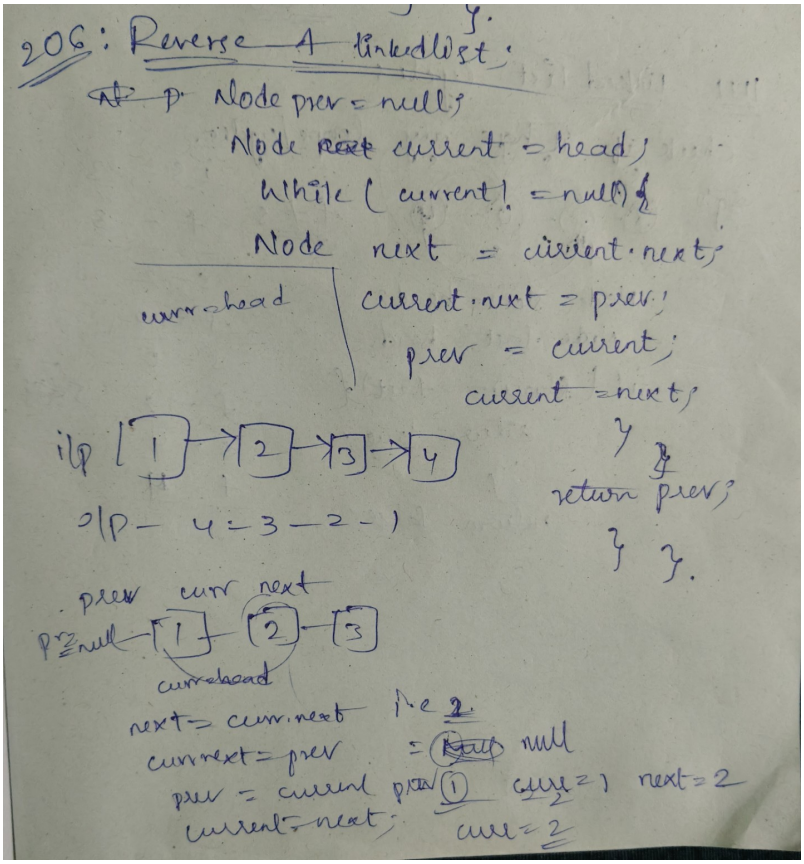
https://leetcode.com/problems/reverse-linked-list/

Problem List

ode

Auto

```
 *     ListNode next;
 *     ListNode() {}
 *     ListNode(int val) { this.val = val; }
 *     ListNode(int val, ListNode next) { this.val = val; this.next =
 * }
 */
class Solution {
    public ListNode reverseList(ListNode head) {
        ListNode prev = null;
        ListNode curr = head;
        while(curr!= null){
        ListNode next = curr.next;
         curr.next = prev;
         prev = curr;
         curr = next;
        }
         return prev;
    }
}
```



17. Middle of the Linked List

https://leetcode.com/problems/middle-of-the-linked-list/

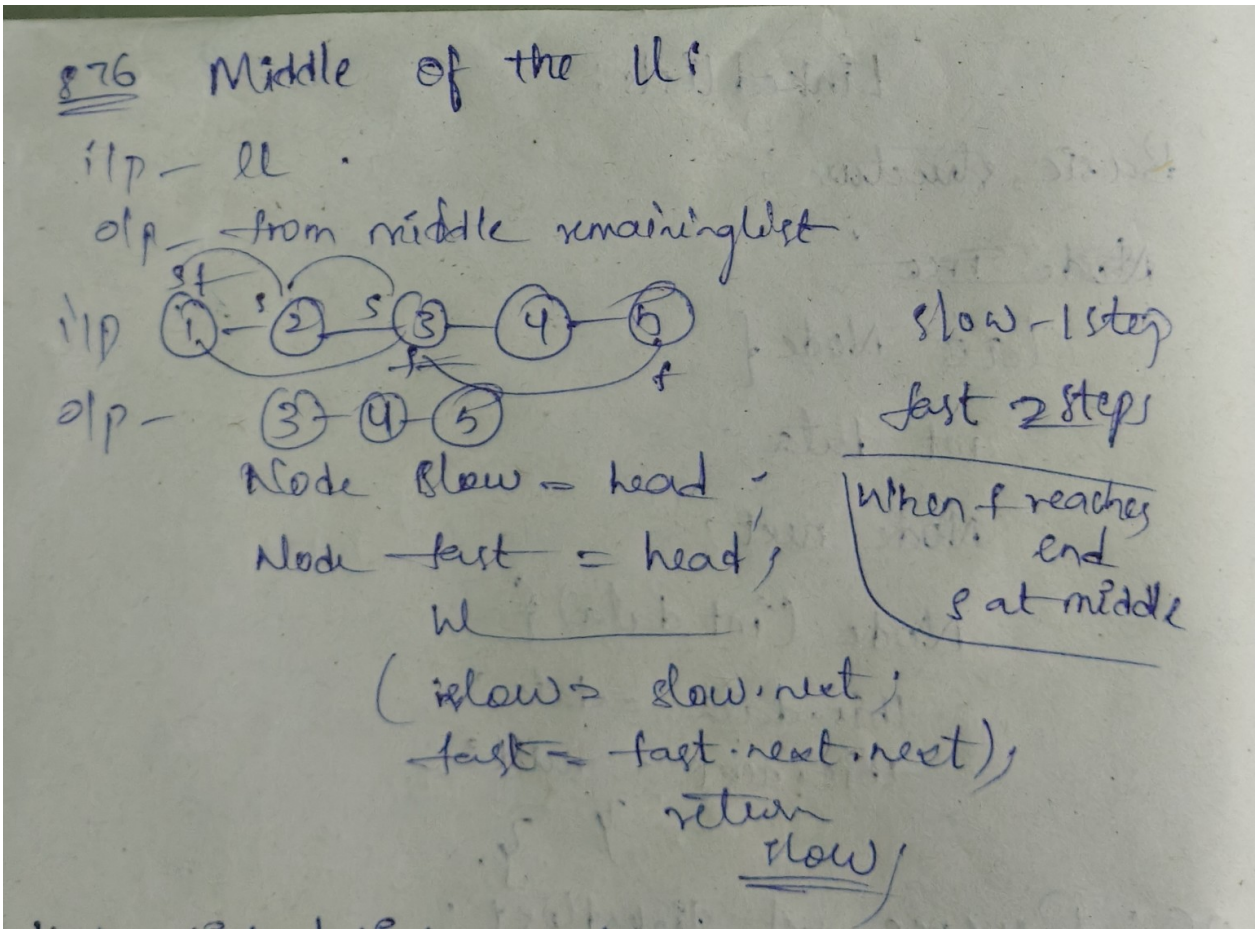
Problem List

Code

Java Auto

```
/*
 * public class ListNode {
 *     int val;
 *     ListNode next;
 *     ListNode() {}
 *     ListNode(int val) { this.val = val; }
 *     ListNode(int val, ListNode next) { this.val = val; this.next = next; }
 * }
 */
class Solution {
    public ListNode middleNode(ListNode head) {
    ListNode slow = head;
     ListNode fast = head;
     while(fast!=null && fast.next!=null){
     slow = slow.next;
     fast = fast.next.next;
     }
     return slow;
    }
}
```

Saved



18. Linked List Cycle

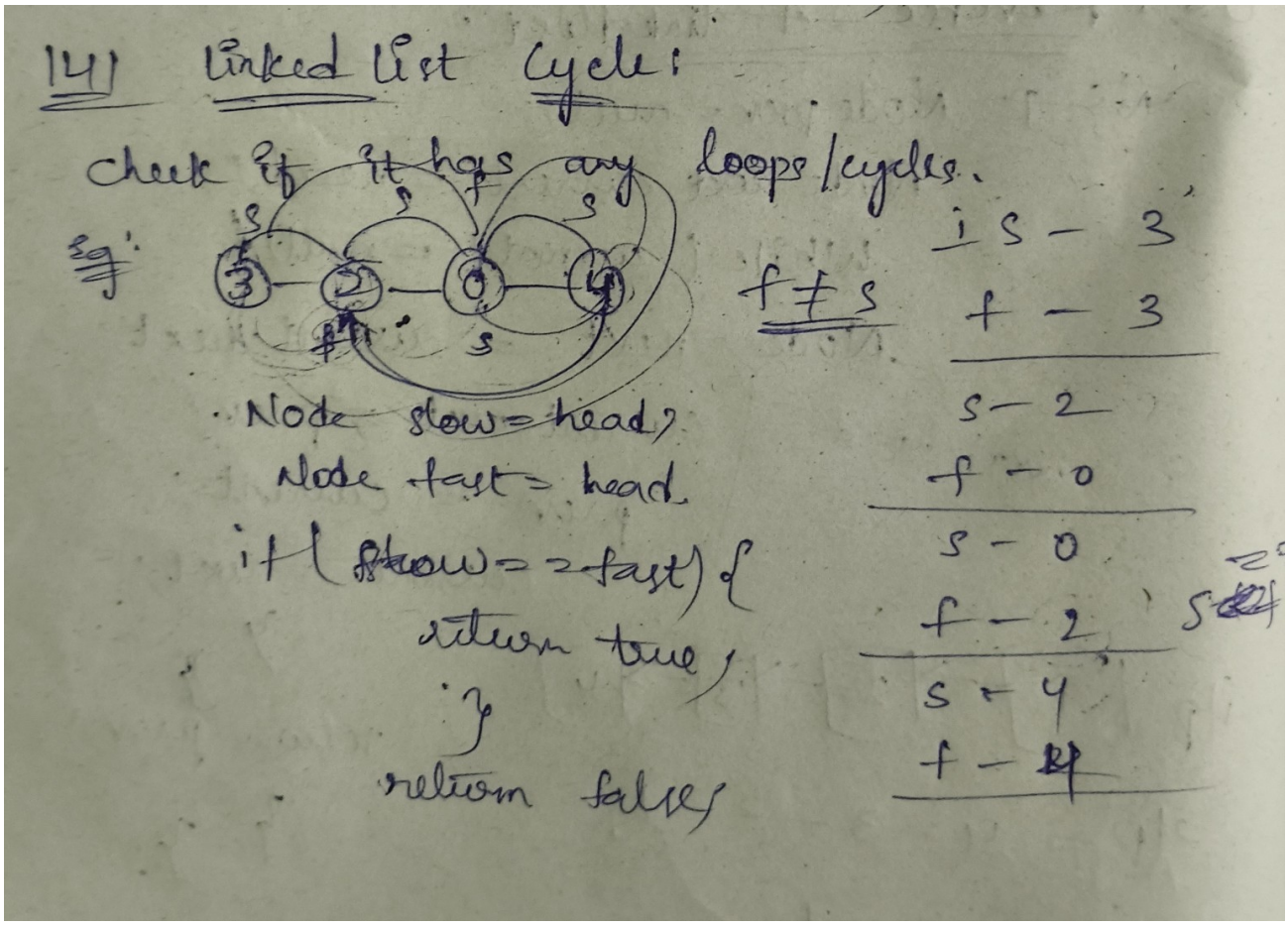
https://leetcode.com/problems/linked-list-cycle/

Problem List

Code

va Auto

```java
/**
 * class ListNode {
 *     int val;
 *     ListNode next;
 *     ListNode(int x) {
 *         val = x;
 *         next = null;
 *     }
 * }
 */
public class Solution {
    public boolean hasCycle(ListNode head) {
        ListNode slow = head;
        ListNode fast = head;
        while(fast!= null && fast.next!=null){
            slow = slow.next;
            fast = fast.next.next;
        if(slow==fast){
        return true;
                }
            }
        return false;
    }
}
```



19. Merge Two Sorted Lists

https://leetcode.com/problems/merge-two-sorted-lists/

Problem List

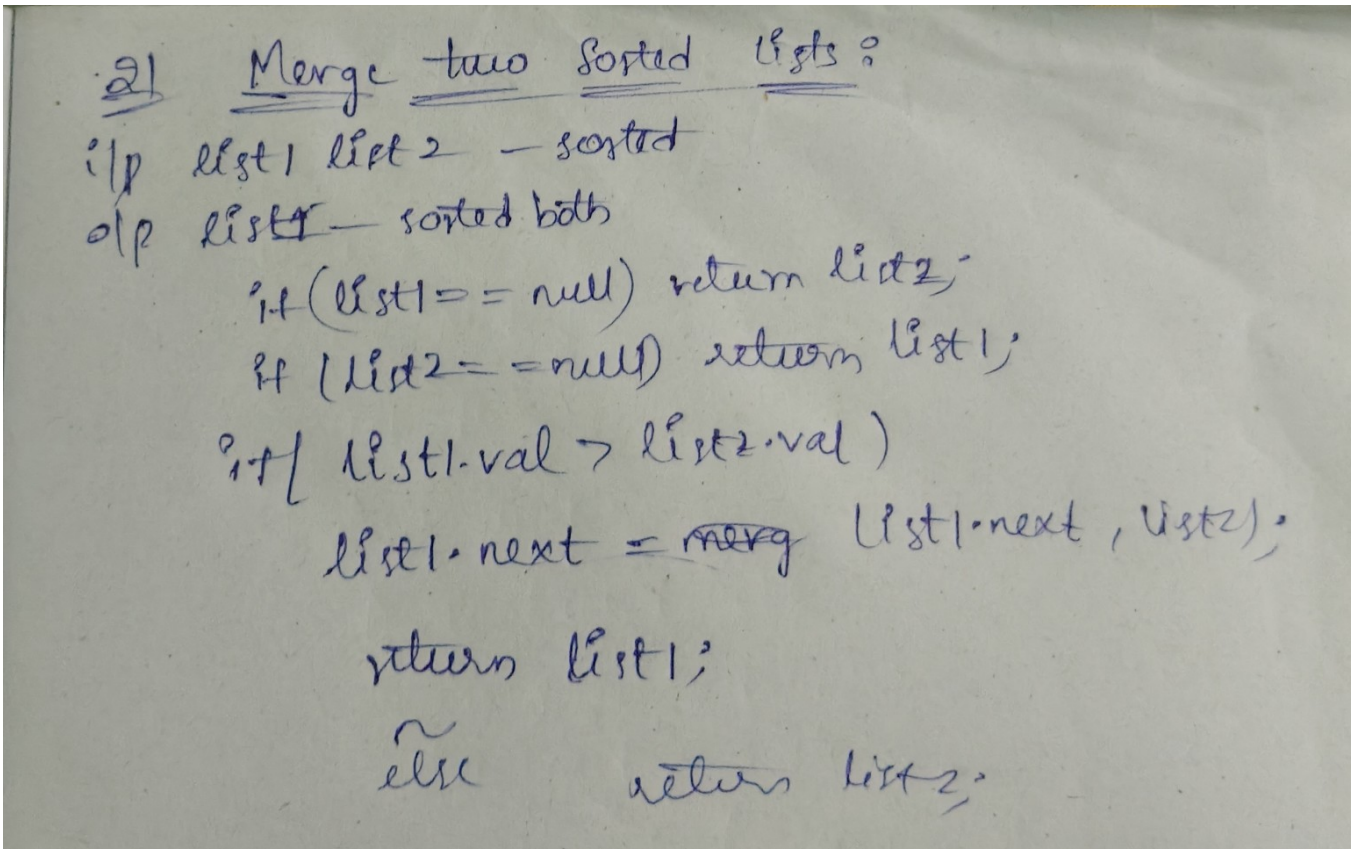
Submit

Code

va Auto

```
/**
 * public class ListNode {
 *     int val;
 *     ListNode next;
 *     ListNode() {}
 *     ListNode(int val) { this.val = val; }
 *     ListNode(int val, ListNode next) { this.val = val; this.next = next; }
 * }
 */
class Solution {
    public ListNode mergeTwoLists(ListNode list1, ListNode list2) {
        List1 curr1 = head;
        List1 curr2 = head;
        while(curr1!=null && curr2!=null){
         List1 next = curr1.next;
         List2 next = curr2.next;
         if(curr1.val > curr2.val){

         }
        }
    }
}
```



20. Remove Nth Node From End of List

https://leetcode.com/problems/remove-nth-node-from-end-of-list/

Problem List

Submit

Code

Auto

```
*     ListNode(int val, ListNode next) { this.val = val; this.next = next; }
* }
*/
class Solution {
    public ListNode removeNthFromEnd(ListNode head, int n) {
        int count = 0;
        ListNode temp = head;
        while(temp != null) {
            count++;
            temp = temp.next;
        }
        if(count == n) {
            return head.next;
        }
        temp = head;
        for(int i = 1; i < count - n; i++) {
            temp = temp.next;
        }
        temp.next = temp.next.next;
        return head;
    }
}
```

